Functions - Part 1: Intro, Domain & Range and Graphs

Course: Pre-Calculus-I

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What is a Function?

Definition

A **function** f from a set A to a set B is a rule that assigns to each element $x \in A$ **exactly one** element $y \in B$.

$$f : A \to B, \quad y = f(x)$$

- x is the **input** (independent variable).
- y is the **output** (dependent variable).
- The set A is the **Domain**.
- The set B is the **Codomain**.
- The set of all actual outputs is the **Range**.

Key Idea: One input $\to$ Exactly one output.

Examples of Functions

- $f(x) = 2x + 3$ (Linear)
- $g(x) = x^2 - 4$ (Quadratic)
- $h(x) = \sqrt{x}$ (Square Root)
- $k(x) = \frac{1}{x}$ (Rational)
- $p(x) = \sin(x)$ (Trigonometric)

Non-Example

The equation $x^2 + y^2 = 25$ is **not** a function because for $x = 3$, y can be 4 or -4.

Domain and Range: Definitions

Domain

The **Domain** of a function is the set of all possible input values (x-values) for which the function is defined.

Range

The **Range** of a function is the set of all possible output values (y-values) that the function can produce.

Notation:

- Interval notation: (a, b), $[a, b)$, $(-\infty, \infty)$
- Set-builder notation: $\{x \mid x > 0\}$

How to Find Domain and Range

Steps to find the Domain:

1. Start with all real numbers: $\mathbb{R}$.
2. Exclude values that cause:
   - Division by zero.
   - Even roots of negative numbers.
   - Logarithms of non-positive numbers.

Steps to find the Range:

1. Solve $y = f(x)$ for x in terms of y.
2. Determine which y-values give a valid x in the domain.
3. Or, analyze the graph of the function.

Example 1: Domain and Range of a Linear Function

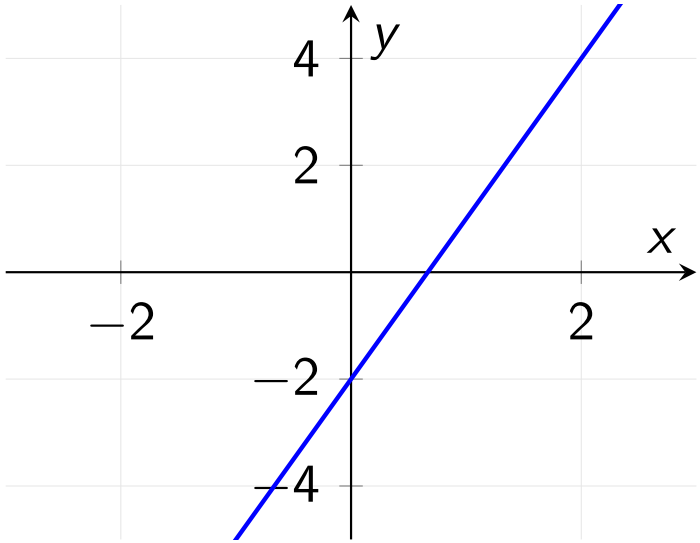
Find the domain and range of $f(x) = 3x - 2$.

Domain:

- No restrictions (no division, no roots).
- Domain$(f) = \mathbb{R} = (-\infty, \infty)$.

Range:

- As x varies over all reals, $3x - 2$ also takes all real values.
- Range$(f) = \mathbb{R} = (-\infty, \infty)$.



Example 2: Domain and Range of a Quadratic Function

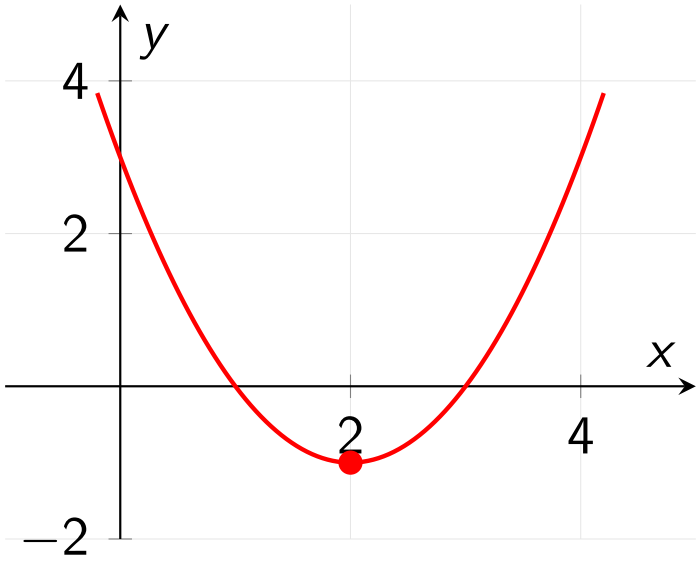
Find the domain and range of $f(x) = x^2 - 4x + 3$.

Domain:

- Polynomial, no restrictions.
- Domain$(f) = \mathbb{R}$.

Range:

- Vertex at $x = -\frac{b}{2a} = 2$.
- $f(2) = 4 - 8 + 3 = -1$.
- Parabola opens upward.
- Range$(f) = [-1, \infty)$.



Example 3: Domain and Range of a Square Root Function

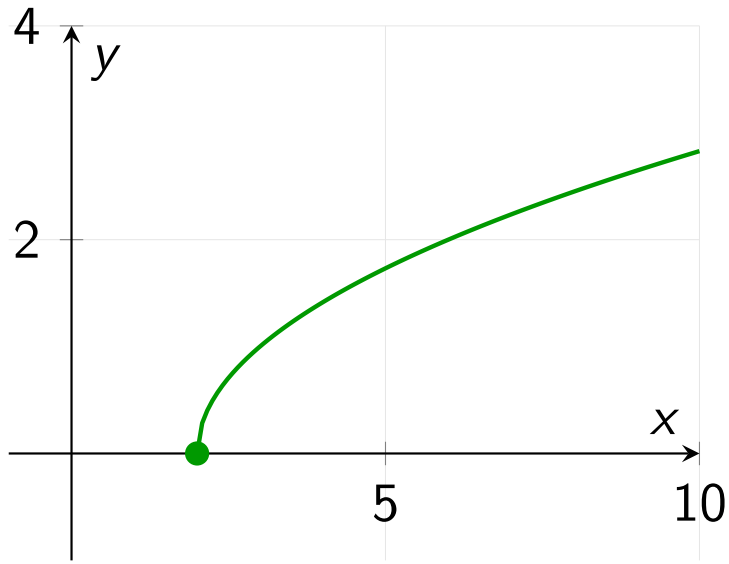
Find the domain and range of $f(x) = \sqrt{x-2}$.

Domain:

- Need $x - 2 \geq 0$.
- $x \geq 2$.
- Domain$(f) = [2, \infty)$.

Range:

- Principal square root is ≥ 0.
- As $x \to \infty$, $\sqrt{x-2} \to \infty$.
- Range$(f) = [0, \infty)$.

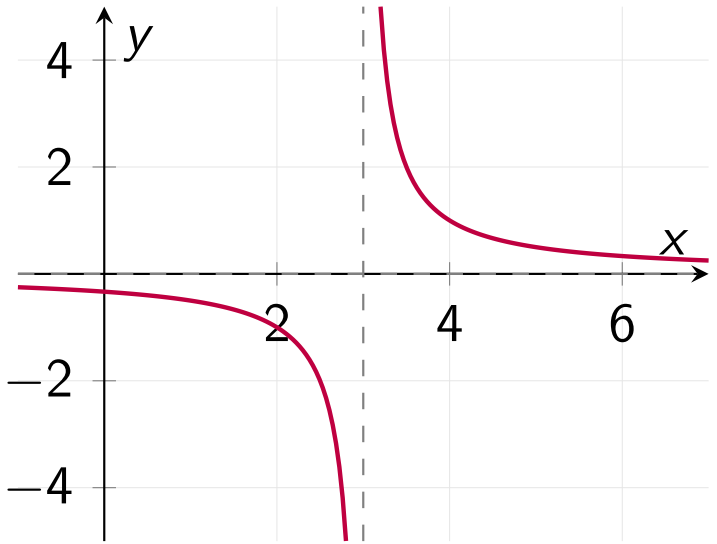


Example 4: Domain and Range of a Rational Function

Find the domain and range of $f(x) = \frac{1}{x-3}$.

Domain:

- Denominator cannot be zero: $x - 3 \neq 0$.
- $x \neq 3$.
- Domain$(f) = \mathbb{R} \setminus \{3\} = (-\infty, 3) \cup (3, \infty)$.



Range:

- $y = \frac{1}{x-3} \implies x = 3 + \frac{1}{y}$.
- $y \neq 0$ (otherwise $1/y$ undefined).
- Range$(f) = \mathbb{R} \setminus \{0\}$.

Example 5: Domain and Range of a Trigonometric Function

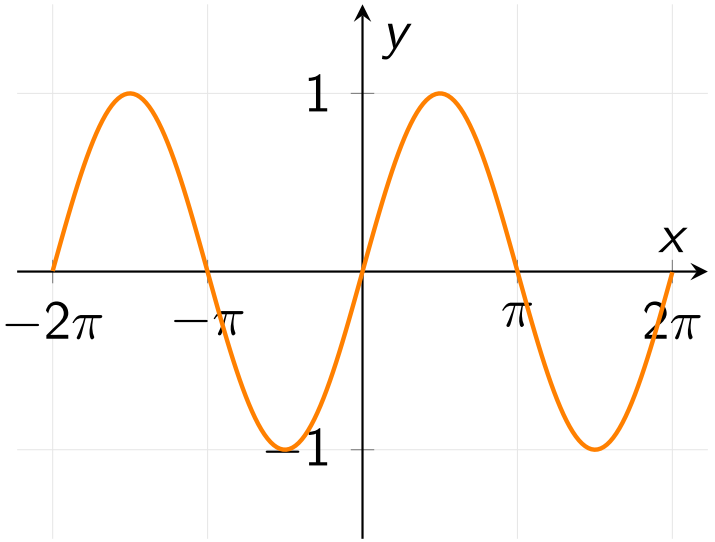
Find the domain and range of $f(x) = \sin(x)$.

Domain:

- Sine is defined for all real numbers.
- $\text{Domain}(f) = \mathbb{R} = (-\infty, \infty)$.

Range:

- The sine function oscillates between -1 and 1.
- $\text{Range}(f) = [-1, 1]$.



What is the Graph of a Function?

Definition

The **graph** of a function f is the set of all ordered pairs (x, y) such that $y = f(x)$ and x is in the domain of f.

$$\text{Graph}(f) = \{(x, f(x)) \mid x \in \text{Domain}(f)\}$$

Key Points:

- It is a visual representation of the function.
- The x-axis represents the input (domain).
- The y-axis represents the output (range).
- **Vertical Line Test:** A curve is a graph of a function if and only if no vertical line intersects the curve more than once.

How to Draw the Graph of a Function

Step-by-Step Method:

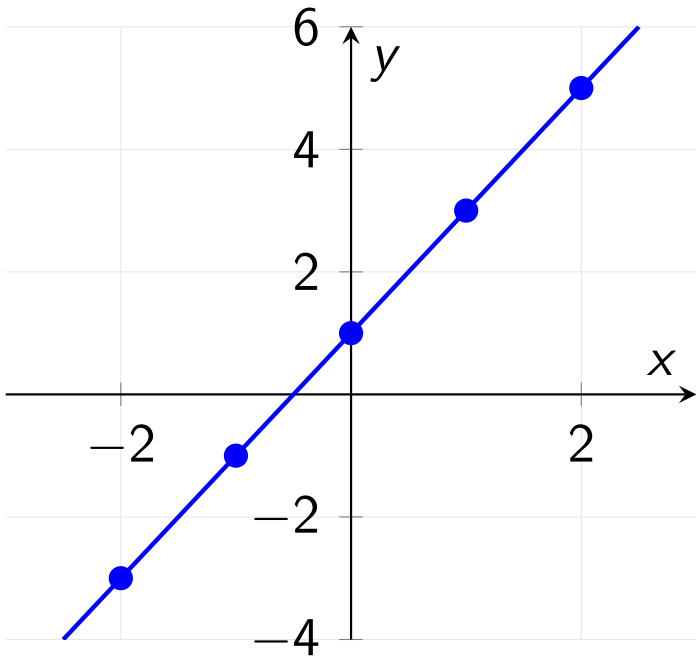
1. **Find the Domain:** Determine where the function is defined.
2. **Find Intercepts:**
   - y-intercept: Set $x = 0$, find $f(0)$.
   - x-intercepts: Set $f(x) = 0$, solve for x.
3. **Create a Table of Values:** Choose several x-values and compute corresponding y-values.
4. **Plot the Points:** Plot the ordered pairs on the Cartesian plane.
5. **Connect the Points:** Draw a smooth curve through the points.
6. **Check for Asymptotes and Symmetry:**
   - Vertical/horizontal asymptotes.
   - Even/Odd symmetry.

Example 1: Graph of a Linear Function

Draw the graph of $f(x) = 2x + 1$.

Table of Values:

x	$y = 2x + 1$
-2	-3
-1	-1
0	1
1	3
2	5



Intercepts:

- y-intercept: $(0, 1)$
- x-intercept: $(-0.5, 0)$

Example 2: Graph of a Quadratic Function

Draw the graph of $f(x) = x^2 - 2x - 3$.

Key Features:

- Vertex: $x = 1$, $y = -4$. Point: $(1, -4)$.
- y-intercept: $(0, -3)$.
- x-intercepts: $(3, 0)$ and $(-1, 0)$.
- Opens upward.

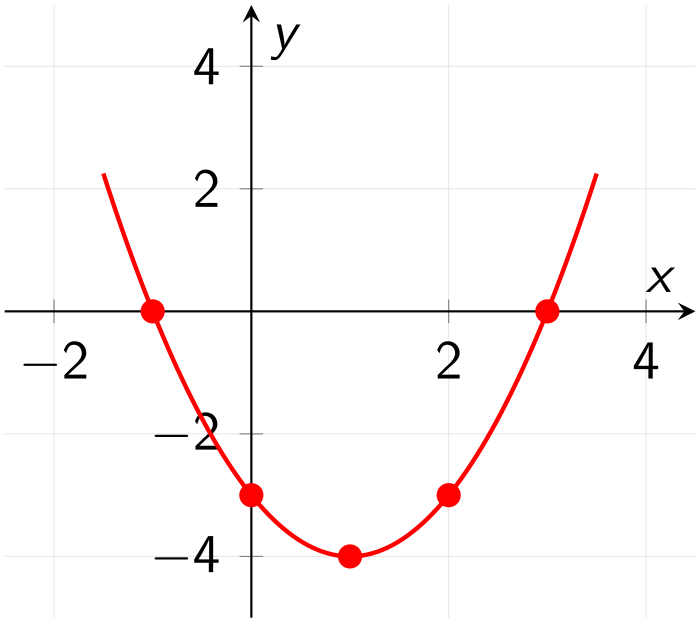


Table of Values:

x	y
-1	0
0	-3
1	-4
2	-3
3	0

Example 3: Graph of a Cubic Function

Draw the graph of $f(x) = x^3 - 3x$.

Key Features:

- Odd symmetry: $f(-x) = -f(x)$.
- y-intercept: $(0, 0)$.
- x-intercepts: $x(x^2 - 3) = 0 \implies x = 0, \pm\sqrt{3} \approx \pm 1.73$.
- Local max at $(-1, 2)$, local min at $(1, -2)$.

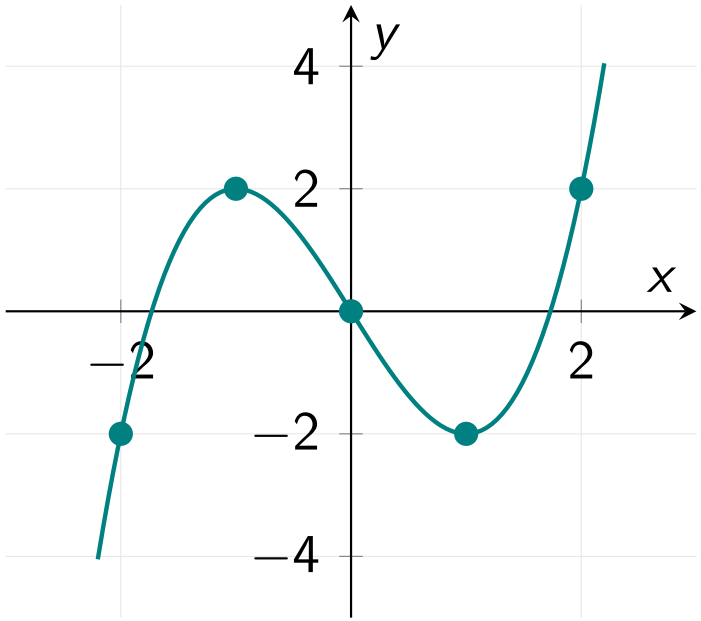


Table of Values:

x	y
-2	-2
-1	2
0	0

Example 4: Graph of an Absolute Value Function

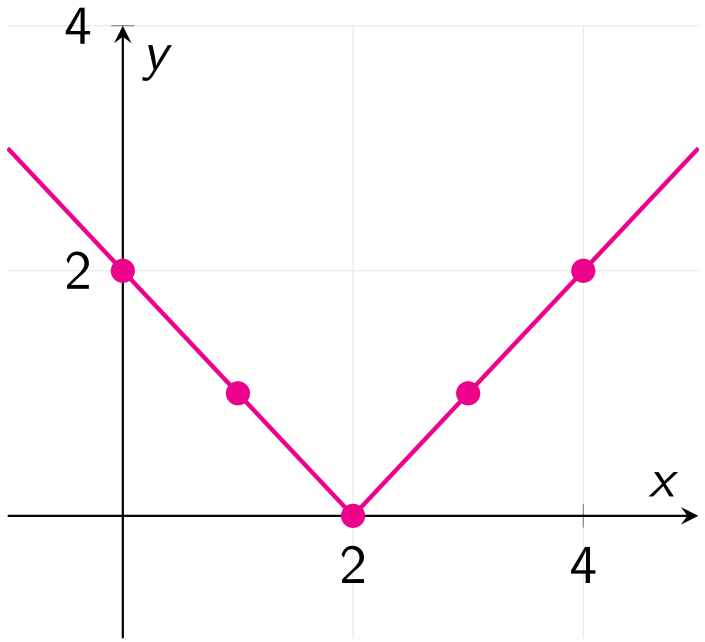
Draw the graph of $f(x) = |x - 2|$.

Key Features:

- Vertex at $(2, 0)$.
- y-intercept: $(0, 2)$.
- x-intercept: $(2, 0)$.
- V-shaped graph.

Table of Values:

x	$y = \lvert x - 2 \rvert$
0	2
1	1
2	0
3	1
4	2



Example 5: Graph of a Rational Function

Draw the graph of $f(x) = \frac{1}{x}$.

Key Features:

- Domain: $x \neq 0$.
- Range: $y \neq 0$.
- Vertical asymptote: $x = 0$.
- Horizontal asymptote: $y = 0$.
- Odd symmetry.

Table of Values:

x	$y = 1/x$
-2	-0.5
-1	-1
-0.5	-2
0.5	2
1	1

